Local arc transitivity from small stabilizers

Human Compression Project

Abstract

We determine the local arc lengths of the configurations in Tables 1–6 of Buekenhout, De Saedeleer and Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II.* The proof uses orders of arc stabilizers, kernels of small permutation actions, and one argument for affine subfield covers. Two configurations with stabilizers S_4 require a longer calculation; the same elementary facts about Sylow two-subgroups settle both. These results concern the specified acting group. They do not determine the full automorphism group of an incidence graph.

1 Arcs and stabilizers

Let X be a connected bipartite graph with vertex types $0, 1$, and let G act faithfully, preserve types, and be transitive on edges. Suppose the degrees d_0, d_1 are at least two. An s-arc is a sequence $(x_0, \dots, x_s)$ of adjacent vertices with $x_{i-1} \neq x_{i+1}$. The action is locally s-arc-transitive if G_x is transitive on the s-arcs starting at x, for each vertex x. We assume local two-transitivity, equivalently local 2-arc-transitivity in this setting. Write K_x for the kernel of G_x on $\mathcal{N}(x)$.

Lemma 1. *The number of s-arcs starting at a vertex of type i is*

$$a_i(s) = d_i \prod_{j=1}^{s-1} (d_{i+j \bmod 2} - 1).$$

Local s-arc-transitivity implies $a_i(s) \mid |G_x|$ for both types. Given local s-arc-transitivity, extension to $s+1$ is equivalent to transitivity of $G_{x_0 \cdots x_s}$ on $\mathcal{N}(x_s) \setminus \{x_{s-1}\}$ for each of the two starting types.

Proof. There are d_i choices for the first edge and, at each later step, all neighbors except the preceding vertex are allowed. This proves the product formula. Orbit–stabilizer gives divisibility. To compare two longer arcs, first carry their initial s-arcs to one another; the remaining possible maps are precisely the stabilizer of that initial arc. This proves the last assertion, including its converse. □

In particular, the stabilizer of a 2-arc (x, y, z) is a pointwise two-point stabilizer in the action of G_y on its neighbors. Its order is $|G_y|/[d_y(d_y - 1)]$. This observation uses the middle vertex; confusing it with an endpoint would give the wrong order.

Lemma 2. *If $d_1 = 2$, the graph is locally 3-arc-transitive.*

Proof. For a 2-arc (x, y, z) with y of type 1, the subgroup fixing x inside G_y fixes both neighbors of y, since there are only two. Thus $G_{xyz} = G_{yz}$. Local two-transitivity at z says that this point stabilizer is transitive on $\mathcal{N}(z) \setminus \{y\}$. For the other starting type there is only one possible extension. Apply Lemma 1. □

For a coset graph with parabolics M, L and intersection J, the degrees are $[M : J], [L : J]$. Connectedness is equivalent to $\langle M, L\rangle = G$. The action is locally two-transitive exactly when the actions of M, L on their cosets of J are two-transitive. These facts follow directly by translating an incident pair to the standard pair of cosets. In the applications, the source's retained constructions supply the subgroups and these local actions. The argument below does not use the source's exhaustive classification, which requires a correction elsewhere.

2 Cases settled by orders and prime cycles

We use D_{2m} for the dihedral group of order $2m$. The following table gives all cases with equal parabolic types in which the obstruction to a third arc is immediate.

$M = L$	J	$d_0 = d_1$	$a_0(3)$
A_5	D_{10}	6	150
A_5	A_4	5	80
A_4	C_3	4	36
S_4	S_3	4	36
S_3	C_2	3	12

Each last entry exceeds the parabolic order, so the exact local arc length is two. Reversing the types has no effect on that conclusion.

For degree two at one type, Lemma 2 proves a third arc. The following obstructions prove that the exact length is three:

M	L	J	d_0	$a_1(4) = 2(d_0 - 1)^2$	$\lvert L\rvert$
A_5	D_{20}	D_{10}	6	50	20
A_5	S_4	A_4	5	32	24
A_4	C_6 or S_3	C_3	4	18	6
S_4	D_{12}	S_3	4	18	12
S_3	C_2^2	C_2	3	8	4

Proposition 3. *The configurations* (A_5, D_{30}, D_{10}) *and* (S_4, D_{18}, S_3) *have exact local arc length three.*

Proof. Treat the two cases together, with $p = 5$ and $p = 3$, respectively. The degrees are $p + 1, 3$. The action of D_{6p} on its three cosets of D_{2p} has kernel the rotation subgroup C_p and quotient S_3. Indeed, writing $D_{6p} = \langle r, t\rangle$ with $|r| = 3p$, the subgroup is $\langle r^3, t\rangle$, and the quotient by $\langle r^3\rangle$ is the usual degree-three dihedral action.

A 2-arc ending at the first type therefore has stabilizer C_p. Its local action is nontrivial: the degree-six action of A_5 is faithful by simplicity, and the natural degree-four action of S_4 is faithful. On the p available next vertices a nontrivial group of order p is transitive. A 2-arc ending at the second type has stabilizer of order two, since the pointwise two-point stabilizers in these two actions of M have order two. It cannot lie in the kernel C_p, so it exchanges the two available next vertices. This proves local 3-arc-transitivity in both orientations. Finally $a_0(4) = 4p(p + 1) > 2p(p + 1) = |M|$, which excludes a fourth arc. □

3 Affine covers of a subfield point stabilizer

Let $K = \mathbb{F}_s$, with $s \geq 2$, and let V be a vector space of dimension two over K, with a fixed line W. Consider the local configuration

$$M = \mathrm{PGL}(2, K), \qquad L = V \rtimes K^*, \qquad J = W \rtimes K^*.$$

The embedding of J in M is its usual point stabilizer on $\mathbb{P}^1(K)$. In characteristic two $M = \mathrm{PSL}(2, K)$. The argument only uses these embedded local actions, so it also applies when V is a two-dimensional subspace of a larger coefficient field.

Proposition 4. *Every such coset configuration has exact local arc length three.*

Proof. The action of M has degree $s + 1$. The action of L on its cosets of J identifies with the affine action on V/W, of degree s, whose kernel is W. The quotient is $\mathrm{AGL}(1, K)$, which is sharply two-transitive.

For a 2-arc ending at an M-vertex, the middle vertex has stabilizer L, so the arc stabilizer is exactly W. Inside the endpoint group M, this is the translation subgroup of a point stabilizer. It is regular on the s other points. In the reverse orientation the arc stabilizer is a two-point stabilizer of $\mathrm{PGL}(2, K)$, of order $s-1$. It lies in J and has trivial intersection with the translation group W: a nonidentity translation fixes only the point at infinity. It therefore projects onto the full multiplicative group of the affine quotient L/W and is transitive on the $s - 1$ available next vertices. This includes $s = 2$, when that last set has one element.

Thus both orientations extend to three arcs. At an L-vertex the number of 4-arcs is $s^3(s-1)$, whereas $|L| = s^2(s - 1)$. Lemma 1 excludes local 4-arc-transitivity. □

This applies to the $A_5, E_{16} : 3, A_4$ configurations in Table 1, to the even-characteristic subfield row of Table 4 where it exists, and to the $\mathrm{PGL}(2, p^n)$ affine row of Table 6. It also proves the same local value for the missing two-dimensional scalar covers in the corrected even-characteristic classification. Their existence and orbit counts are separate assertions and are not inferred from this local proof.

4 The two-subgroup calculations in S_4

We record the elementary group facts used below. In its action of degree three on partitions into two pairs, S_4 has kernel the normal four-group

$$V = \{1, (12)(34), (13)(24), (14)(23)\}.$$

The three point stabilizers are its Sylow two-subgroups, each isomorphic to D_8. Distinct such subgroups intersect in V. Their three central involutions are the three distinct nonidentity elements of V. These statements follow by taking the stabilizer of $\{\{1, 2\}, \{3, 4\}\}$ and conjugating it to the other two partitions.

In $D_8 = \langle a, b \mid a^4 = b^2 = 1,\ bab = a^{-1} \rangle$ there are exactly two four-groups, $\langle a^2, b \rangle$ and $\langle a^2, ab \rangle$. They intersect in the center $\langle a^2 \rangle$. The unique cyclic subgroup of order four intersects either in that same center. If this D_8 has index two in D_{16}, conjugation by an element outside it interchanges its two four-groups. To see this, write $D_{16} = \langle r, b \rangle$, $|r| = 8$, and $a = r^2$; conjugation by r sends one of the two displayed four-groups to the other.

Proposition 5. *The configuration (S_4, D_{24}, D_8) has exact local arc length three.*

Proof. Both degrees are three. At an S_4-vertex the local kernel is V. At a D_{24}-vertex it is C_4: in the presentation with rotation r of order twelve, the subgroup D_8 is $\langle r^3, b\rangle$, and quotienting by $\langle r^3\rangle$ gives S_3. Since both local quotients are sharply two-transitive of degree three, the stabilizer of a 2-arc is the local kernel of its middle vertex. Inside the endpoint edge stabilizer D_8, neither C_4 nor V is contained in the other. Its nontrivial local image fixes the incoming neighbor and interchanges the other two. This proves the third arc in both orientations.

Choose a 3-arc (a, b, c, d) with a, c of type S_4. The stabilizer of (a, b, c) is C_4 and the stabilizer of the 3-arc is its subgroup of order two. Its generator is a square of an element of order four in $G_a = S_4$, hence is a double transposition in the normal four-group K_a. It fixes every neighbor of a. For the reversed 3-arc (d, c, b, a) it consequently cannot exchange the two possible extensions. Local 4-arc-transitivity fails. □

Proposition 6. *Suppose G is nonabelian simple and is generated by the two parabolics in a configuration (S_4, S_4, D_8). Its exact local arc length is four.*

Proof. Write $V_x = K_x$ at each vertex. If x, y are adjacent and $V_x = V_y$, this nontrivial four-group is normal in both G_x and G_y, and hence in G. This contradicts simplicity. Thus V_x, V_y are the two distinct four-groups in $J_{xy} = G_x \cap G_y$, and

$$V_x \cap V_y = Z(J_{xy}) = \langle z_{xy}\rangle.$$

For a 2-arc (a, b, c), its stabilizer is V_b. Its image on the neighbors of c has order two, since $V_b \cap V_c$ has order two. That image fixes b and interchanges the other two neighbors. The stabilizer of a resulting 3-arc (a, b, c, d) is $V_b \cap V_c = \langle z_{bc}\rangle$.

The subgroups J_{bc} and J_{cd} are different Sylow two-subgroups of $G_c = S_4$, so $z_{bc} \neq z_{cd}$. Both belong to V_c, whereas $V_c \cap V_d = \langle z_{cd}\rangle$. Thus z_{bc} does not belong to the local kernel V_d. It fixes c and exchanges the other two neighbors of d, proving the fourth arc. Finally, $a_i(5) = 48 > 24$, so a fifth arc is impossible for this specified group. □

Proposition 7. *The configuration (S_4, D_{16}, D_8) has exact local arc length seven.*

Proof. Write an alternating path as

$$(v_0, e_0, v_1, e_1, v_2, e_2, v_3, e_3),$$

where the v_i have stabilizer S_4 and the e_i have stabilizer D_{16}. A vertex e_i has just two neighbors. An element fixing e_i and one of them fixes the other; hence the two edge stabilizers there coincide. Write this common group as $J_i \cong D_8$. Let V_i be the local four-group at v_i.

The index-two subgroup J_i is normal in $G_{e_i} = D_{16}$. An element of $G_{e_i} \setminus J_i$ exchanges its endpoints and interchanges the two four-groups in J_i, by the calculation above. Consequently

$$V_i \neq V_{i+1}, \qquad V_i \cap V_{i+1} = Z(J_i) = \langle z_i\rangle.$$

After the first two edges the stabilizer is J_0, of order eight. At v_1 it fixes the incoming neighbor and exchanges the other two, since its kernel there is V_1. The stabilizer after three edges is $J_0 \cap J_1 = V_1$, and the fourth edge is forced.

At v_2 the image of V_1 has order two, since $V_1 \cap V_2 = \langle z_1\rangle$. It fixes the incoming neighbor and exchanges the other two. Thus the fifth step is transitive and its stabilizer is $\langle z_1\rangle$. The sixth step is forced. At v_2, the two different Sylow subgroups J_1, J_2 have different centers, so $z_1 \neq z_2$. Since

$V_2 \cap V_3 = \langle z_2 \rangle$, the element $z_1 \in V_2$ is outside V_3. It exchanges the two continuations at v_3. This proves transitivity on all 7-arcs starting at the S_4 type.

This also gives the other starting type. Vertex transitivity on the S_4 half and the proved local transitivity imply that G is transitive on all 7-arcs starting in that half. Reversal is a G-equivariant bijection to the 7-arcs starting in the other half, since seven is odd. Hence those arcs also form a single G-orbit. Two of them with the same initial vertex can only be mapped to one another by its stabilizer, which proves the required local transitivity. At a degree-two vertex, $a_1(8) = 32 > |D_{16}| = 16$, which excludes an eighth arc. □

5 Comparison with the fixed source

The following is the complete assignment of the local arc column. It does not assert that all accompanying existence or equivalence counts in the source tables are correct; those are separate source obligations.

Source table	Rows	Exact length	Argument
1	$7, 8, 10, 11, 14, 15, 17, 18$	2	equal-type order bound
1	$1, 6, 13$	3	Proposition 3
1	$2, 3, 4$	3	Proposition 4, $s = 4$
1	$5, 9, 12, 16$	3	degree two and order bound
2	$1, 2, 3, 4, 5$	3	degree two and order bound
2	$6, 7$	2	equal-type order bound
3	$1, 9$	3	degree two and order bound
3	2	3	Proposition 3
3	$3, 4$	2	equal-type order bound
3	5	7	Proposition 7
3	6	3	Proposition 5
3	$7, 8$	4	Proposition 6
4	1	3	Proposition 4
4	2	2	equal-type order bound
4	$3, 4$	3	degree two and order bound
4	5	3	Proposition 3
5	$1, 2$	2	A_4, A_4, C_3 order bound
6	1	3	Proposition 4
6	2	3	degree two and order bound
6	3	4	Proposition 6

In Section 6 the printed second orientation of the A_5, D_{30} argument repeats types $0, 1, 0$; it must read $1, 0, 1$. In the S_4, D_{24} argument the printed "D_{18} on cosets of D_8" must read D_{24}. Proposition 5 retains the obstruction attributed there to an unnamed referee. The general arc-stabilizer method is standard and is explicitly attributed by the source to Leemans's earlier work. The contribution here is the complete uniform derivation of the table column, especially the two S_4 calculations and their common use of distinct Sylow centers.

The separate assertions on the full automorphism groups of particular graphs on source p. 386 are not settled by this note. An automorphism group containing the specified G can have greater arc transitivity; the $q = 9$ example explicitly illustrates that distinction. No upper bound on such a larger group follows merely because an outer automorphism of G interchanges two subgroup classes. No full-source completion, Lean verification, new open-question answer or compression factor is claimed by this component.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Tables 1–6 and Section 6.

[2] D. Leemans, *Locally s-arc-transitive graphs related to sporadic simple groups*, Journal of Algebra **322** (2009), 882–892; attribution follows the fixed source.